

Parul University			
Academic Year: 2026-2027			
Name of Teacher: ASHWANI CHAUDHARY			
Subject: Artificial Intelligence			
Subject Code: 303105307			
Lesson/Lab Plan No	Lesson/Lab Plan Title	Lesson/Lab Plan Description	Lesson/Lab Plan Date (dd-mm-yyyy)
1	Unit 1- Introduction	Definition of an AI, Major Areas of Artificial Intelligence	11-06-2026
2	Unit 1- Introduction	AI Techniques, History, AI problems	12-06-2026
3	Unit 1- Introduction	Production Systems, Problem characteristics	13-06-2026
4	Unit 1- Introduction	Intelligent Agents, Agent Architecture	18-06-2026
5	Unit 1- Introduction	AI Application (E-Commerce, & Medicine), AI Representation	19-06-2026
6	Unit 1- Introduction	Properties of internal representation, Future scope of AI	25-06-2026
7	Unit 1- Introduction	Issues in the design of search algorithms.	26-06-2026
8	Unit 2- Search techniques	Generate-And-Test, Hill Climbing, Best-First Search	27-06-2026
9	Unit 2- Search techniques	Problem Reduction, Constraint Satisfaction, Means-Ends Analysis. Heuristic search,	02-07-2026
10	Unit 2- Search techniques	Hill Climbing, Best first search, mean and end analysis,	03-07-2026
11	Unit 2- Search techniques	Constraint Satisfaction, A* and AO* Algorithm,	04-07-2026
12	Unit 2- Search techniques	Knowledge Representation: Basic concepts, Knowledge representation Paradigms	09-07-2026
13	Unit 2- Search techniques	Propositional Logic	10-07-2026
14	Unit 2- Search techniques	Inference Rules in Propositional Logic	16-07-2026
15	Unit 2- Search techniques	Knowledge representation using Predicate logic	17-07-2026
16	Unit 2- Search techniques	Predicate Calculus, Predicate and arguments	18-07-2026
17	Unit 2- Search techniques	ISA hierarchy, Frame notation	23-07-2026
18	Unit 2- Search techniques	Resolution, Natural Deduction	24-07-2026
19	Unit 3 - Knowledge Representation	Knowledge Representation and Mappings	25-07-2026
20	Unit 3 - Knowledge Representation	Different Approaches, Issues in knowledge representation	30-07-2026
21	Unit 3 - Knowledge Representation	Predicate Logic	31-07-2026
22	Unit 3 - Knowledge Representation	Representation Simple Facts in Logic, Representing Instance and Isa Relationships	01-08-2026
23	Unit 3 - Knowledge Representation	Computable Functions and Predicates, Resolution	06-08-2026
24	Unit 3 - Knowledge Representation	Propositional Logic: Representation	07-08-2026
25	Unit 3 - Knowledge Representation	Inference, Reasoning Patterns, Resolution	08-08-2026
26	Unit 3 - Knowledge Representation	First-order Logic: Representation	13-08-2026
27	Unit 3 - Knowledge Representation	Inference	14-08-2026
28	Unit 3 - Knowledge Representation	Reasoning Patterns, Resolution	27-08-2026
29	Unit 4- Uncertainty	Non-Monotonic Reasoning, Logics for Non-Monotonic Reasoning	28-08-2026
30	Unit 4- Uncertainty	Forward rules, and Backward rules, Justification based Truth Maintenance Systems	29-08-2026
31	Unit 4- Uncertainty	Semantic Nets Statistical Reasoning, Probability and Bayes' theorem,	03-09-2026
32	Unit 4- Uncertainty	Bayesian Network, Markov Networks, Hidden Markov Model, Basis of Utility Theory, Utility Functions	04-09-2026
33	Unit 5- Fuzzy Sets and Fuzzy Logic	Fuzzy Set Operations, Membership Functions, Fuzzy Logic	05-09-2026
34	Unit 5- Fuzzy Sets and Fuzzy Logic	Hedges, Fuzzy Proposition and Inference Rules	10-09-2026
35	Unit 5- Fuzzy Sets and Fuzzy Logic	Fuzzy Systems	11-09-2026
36	Unit 6- Natural Language Processing	Introduction, Syntactic Processing,	12-09-2026
37	Unit 6- Natural Language Processing	Semantic Analysis,	17-09-2026
38	Unit 6- Natural Language Processing	Discourse and Pragmatic Processing, Spell Checking	18-09-2026
39	Unit 7- Neural Networks and Expert systems	Introduction to neural networks and perception-qualitative Analysis,	18-09-2026
40	Unit 7- Neural Networks and Expert systems	Neural net architecture and applications	19-09-2026

41	Unit 7-Neural Networks and Expert systems	Utilization and functionality, the architecture of the expert system	22-09-2026
42	Unit 7-Neural Networks and Expert systems	knowledge representation, knowledge representation, two case studies on expert systems	25-09-2026
43	Unit 7-Neural Networks and Expert systems	Two case studies Utilization and functionality	26-09-2026
44	Unit 7-Neural Networks and Expert systems	the architecture of the expert system	27-09-2026
45	Unit 7-Neural Networks and Expert systems	knowledge representation with two case studies	28-09-2026